Risk Cashflows and Interest Rate Delta

1. Start with the payments

A linear interest rate derivative can be valued by adding its discounted fixed and floating payments. Receipts are positive and payments negative:

$$\boxed{V = \sum_i C_i D_i^d + \sum_j N_j \alpha_j D_j^d [F_j(\mathbf{D}) + s_j].} \tag{1}$$

Here C_i is a known cash amount, N_j is signed notional, α_j is the accrual year fraction, and s_j is a fixed spread. D^d is the discount factor under the applicable collateral agreement. F_j is the projected floating coupon rate, including any adjustment required by the payment conventions; after fixing, use the known rate.

The vector $\mathbf{D}$ contains all discount factors used for both discounting and rate projection. A market move can change a floating payment's **amount** as well as the factor used to **discount** it.

2. Calculate risk in two steps

Step 1: how does PV respond to each discount factor?

$$R_k = \frac{\partial V}{\partial D_k}, \qquad \mathbf{R} = \nabla_D V. \tag{2}$$

We call R_k a **risk cashflow**. It has units of currency, but it is a sensitivity, not necessarily a contractual payment. It is the amount that translates a small change in D_k into a PV change.

For a fixed payment, $V = CD$ gives $R = C$: risk cashflow equals the actual payment. For a floating payment, use the **product rule**: include both the change in the projected payment and the change in discounting. If the same curve drives both, combine both effects. A coupon that has already fixed behaves as a fixed payment.

Step 2: how does each discount factor respond to a rate? Let $\mathbf{r}$ contain the underlying rates or calibration quotes. The curve Jacobian J records how each discount factor changes when each rate changes:

$$J_{ka} = \frac{\partial D_k}{\partial r_a}, \qquad \boxed{\boldsymbol{\delta} = \nabla_r V = J^\mathsf{T} \mathbf{R}.} \tag{3}$$

This is the chain rule: **rate change → discount factor changes → PV change**. The transpose sums all discount-factor contributions to each rate bucket. J can be calculated by bumping quotes and recalibrating the affected curves. Rates are in decimal units, so $10^{-4}\delta_a$ approximates P&L for a +1 bp move.

3. Why separate the calculation?

Benefits. Label risk cashflow records by warehouse, product type or other attributes, then filter and aggregate them. Once the records and curve responses are available, different risk buckets can be calculated quickly by matrix multiplication, without repeating trade-level differentiation.

Cost. Generating risk cashflows can take time for a large portfolio. Some are constant; others change with the market and need refreshing. Reusing them saves work, but may introduce the approximation discussed next.

4. SOD and live markets: what is updated?

Call start of day (SOD) state 0 and the live market state 1. Write $\mathbf{R}_0, J_0$ for SOD risk cashflows and curve responses, and $\mathbf{R}_1, J_1$ for their live values. Hold the valuation date, trades, fixings, cashflow state and risk coordinates unchanged: only market curves move.

Method	Risk cashflows	Curve response	Rate delta
A	SOD	SOD	$\boldsymbol{\delta}_A = J_0^\mathsf{T}\mathbf{R}_0$
B	SOD	Live	$\boldsymbol{\delta}_B = J_1^\mathsf{T}\mathbf{R}_0$
C	Live	Live	$\boldsymbol{\delta}_C = J_1^\mathsf{T}\mathbf{R}_1$

A keeps the old risk. B updates the curve response. C updates both. C is the benchmark for current-market delta. Define error as calculated delta minus C:

$$E_B = J_1^\mathsf{T}(\mathbf{R}_0 - \mathbf{R}_1), \qquad E_A = (J_0 - J_1)^\mathsf{T}\mathbf{R}_0 + E_B. \tag{4}$$

B's error comes only from old risk cashflows. A also has an old curve-response contribution. These two contributions can reinforce or offset one another.

Fixed payments. Their risk cashflows do not change, so **B is as good as C**. The same holds whenever PV is **linear plus a constant** in the chosen discount factors (also called *affine*). Floating payments can also have this property, as Example 1 shows.

5. Which approximation is better, A or B?

Here "better" means closer to C, measured by absolute delta error in a particular bucket. Updating only J changes the result by

$$\boxed{\boldsymbol{\delta}_B - \boldsymbol{\delta}_A = (J_1 - J_0)^\mathsf{T}\mathbf{R}_0.} \tag{5}$$

The intuition. True delta can move because risk cashflows change and because curve responses change. B captures the second effect but misses the first. If the effects reinforce one another, B generally helps. If they offset, A can remain close to the live answer, while B breaks the offset.

For a small move in one rate, write the two contributions as

$$\underbrace{u = J_0^\mathsf{T}\Delta\mathbf{R}}_{\text{risk-cashflow effect}}, \qquad \underbrace{v = (\Delta J)^\mathsf{T}\mathbf{R}_0}_{\text{curve-response effect}}. \tag{6}$$

Then $\delta_C - \delta_A \approx u + v$, so

$$E_A \approx -(u+v), \qquad E_B \approx -u. \tag{7}$$

When B is better. To first order, $|u| < |u+v|$. This holds when the two effects have the same sign and $v \neq 0$. B is exact if risk cashflows stay constant.

When A is better. To first order, $|u+v| < |u|$. The effects must oppose, with $0 < |v| < 2|u|$. It is their *effects on delta* that oppose; R and J need not move in opposite numerical directions.

Link to gamma. The two contributions are approximately the market rate move times two parts of gamma: $u \approx \Gamma_{\text{trade}}\Delta r$ and $v \approx \Gamma_{\text{curve}}\Delta r$. The first describes changing risk cashflows; the second describes changing curve responses. Together they give the change in true delta. The next page gives exact numerical examples, rather than relying on this small-move approximation.

Section 5 continued: two floating-coupon examples

Both examples use an unfixed one-year floating coupon with notional $N = 100$, accrual fraction one and no principal repayment. Rates are simple rates. Deltas are per unit decimal rate. Each example specifies both the market move and the rate used to measure risk.

Example 1: B is better when risk cashflows stay constant. Use one rate r for both projection and discounting. With $D = 1/(1 + r)$,

$$V = NrD = N(1 - D), \qquad R = -N, \qquad J = -\frac{1}{(1 + r)^2}. \tag{8}$$

Move r from 5% to 5.1% and calculate sensitivity to r. The projected coupon amount changes, but its risk cashflow remains -100.

Method	Rate delta	Error versus C
A: both SOD	90.7029	+0.1725
B: SOD risk cashflow, live Jacobian	90.5304	0
C: both live	90.5304	0

Why B wins. All the delta change comes from J. Updating J therefore gives the full live answer, even though the risk cashflow was calculated at SOD.

Example 2: A is better when the two effects cancel. Use separate simple discount and projection rates d and p. Hold $d = 5\%$ fixed, move only p from 5% to 5.1%, and measure sensitivity to p. Let $x = 1/(1 + d)$ and $z = 1/(1 + p)$. Then

$$V = Nx\left(\frac{1}{z} - 1\right) = \frac{Np}{1 + d}, \qquad \frac{\partial V}{\partial p} = \frac{N}{1 + d}. \tag{9}$$

The true projection-rate delta does not change. However, its two building blocks do:

$$R_z = -\frac{Nx}{z^2}, \qquad J_{zp} = -\frac{1}{(1 + p)^2}, \qquad \delta_B = \frac{N}{1 + d}\left(\frac{1 + p_0}{1 + p_1}\right)^2. \tag{10}$$

Here p_0, p_1 are the SOD and live projection rates. The discount factor x stays fixed for this risk direction.

Method	Projection-rate delta	Error versus C
A: both SOD	95.2381	0
B: SOD risk cashflow, live Jacobian	95.0569	−0.1811
C: both live	95.2381	0

Why A wins. As p rises, $|R_z|$ increases and $|J_{zp}|$ decreases. Their product stays constant. A keeps the old, correct product; B updates only one factor and introduces an error. Moving discount and projection rates together would be a different comparison.

Conclusion. Neither A nor B is generally closer to C. B works well when risk cashflows change little; A can be closer when two changes offset. Compare both with C for the relevant portfolio, market moves and individual buckets. A small aggregate error can hide larger bucket errors.